SDG BLOCKCHAIN ACCELERATOR

Impact Measurement Framework

In collaboration with
UNDP Bosnia and Herzegovina / University of Sarajevo) / Sala

1. Project Information

Project Name: Digital Identity & Credential Verification
Challenge & UNDP Office: UNDP Bosnia and Herzegovina
Challenge Owner: University of Sarajevo
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2. SDG Alignment

This project contributes to multiple Sustainable Development Goals by introducing a verifiable and portable digital credential infrastructure built on blockchain technology.

SDG 4 – Quality Education (Primary)

- **Target 4.4:** Increase the number of youth and adults with relevant skills for employment
- **Target 4.a:** Build effective and inclusive education systems

Contribution:
The platform enables trusted, verifiable academic credentials that can be easily shared and verified across institutions and employers, improving employability and academic mobility.

SDG 16 – Peace, Justice and Strong Institutions

- **Target 16.6:** Develop effective, accountable, and transparent institutions
- **Target 16.10:** Ensure public access to information

Contribution:
The system introduces a transparent and tamper-resistant verification process, reducing reliance on manual checks and institutional intermediaries.

SDG 9 – Industry, Innovation and Infrastructure

- **Target 9.1 / 9.c:** Develop resilient and inclusive infrastructure

Contribution:
The solution establishes blockchain-based verification infrastructure that can be extended across institutions and integrated into digital public systems.

SDG 10 – Reduced Inequalities (Enabling)

- **Target 10.2:** Promote social and economic inclusion

Contribution:
By enabling cross-border and independent credential verification, the system reduces barriers for individuals seeking opportunities in different regions.

3. Impact Model

Theory of Change

```
Digital Credential Issuance
        ↓
On-chain Proof Anchoring
        ↓
Instant, Trustless Verification
        ↓
Reduced Fraud and Administrative Burden
        ↓
Improved Access to Education and Employment Opportunities
```

Impact Chain

Level	Description	Measurement
Output	Credentials issued on platform	Backend + blockchain logs
Outcome	Credentials verified by third parties	Verification events
Efficiency	Reduced manual verification workload	Institutional reporting
Impact	Increased trust and accessibility of credentials	Surveys + adoption metrics

4. Key Performance Indicators (KPIs)

4.1 Core System KPIs

KPI	Measurement Method	Target (Pilot)
Credentials issued	Backend + blockchain logs	10–50
Successful verifications	API + QR logs	≥95%
On-chain transaction success	Blockchain logs	≥98%
Verification time	System response time	<3 seconds
Invalid verification rejection rate	Validator logs	100%

4.2 Adoption KPIs

KPI	Measurement	Target
Participating institutions	Number of issuers	1–2
Active users (students)	Platform usage	20–100

Verification requests	QR scans / API calls	100–500

4.3 Efficiency KPIs

KPI	Measurement	Target
Reduction in manual verification	Institutional reporting	≥50%
Verification time improvement	Process comparison	≥70% faster
Cost reduction per verification	Estimated operational cost	≥30%

4.4 Trust & User Experience KPIs

KPI	Measurement	Target
Verifier satisfaction score	Micro-survey	≥80%
Perceived trust level	Survey	≥80%
Error rate	Failed verifications	<5%

5. Measurement & Tracking Methods

5.1 On-Chain Metrics

Captured via blockchain indexers (e.g., Blockfrost):

- Transaction success rate
- Credential minting events
- Validator execution outcomes
- Rejected invalid transactions

5.2 Off-Chain Metrics

Captured via backend services and analytics:

- Credential issuance logs
- Verification requests (API + QR scans)
- User activity and engagement
- System performance metrics

5.3 Institutional Data Collection

- Monthly reporting from participating institutions
- Tracking reduction in manual verification requests
- Feedback on operational efficiency improvements

5.4 Survey-Based Data

Micro-surveys triggered post-verification:

- Verification speed satisfaction
- Trust in credential authenticity
- Likelihood of reuse

6. Baseline vs Projected Outcomes

Metric	Baseline	Pilot Target
Digitally verifiable credentials	0	10–50
Verification requests	0	100–500
Manual verification workload	High	Reduced by ≥50%
Verification success rate	N/A	≥95%
Verification time	1–3 days	<3 seconds

7. Monitoring & Reporting Plan

Frequency

- Weekly internal technical monitoring
- Monthly pilot performance review
- Quarterly reporting to UNDP

Ownership

Role	Responsibility
Technical Lead	On-chain and system metrics

Product / Project Manager	KPI tracking and reporting
UNDP Partner	Institutional validation and oversight

Reporting Outputs

- Live dashboard (system metrics)
- Monthly pilot summary report
- Quarterly impact report for UNDP and stakeholders

8. Risks & Mitigation

Adoption Risk

Risk: Limited participation from institutions
Mitigation: Start with a controlled pilot and expand gradually

Data Consistency Risk

Risk: Mismatch between on-chain and off-chain data
Mitigation: Hash-based validation and periodic reconciliation

User Experience Risk

Risk: Users may struggle with verification flow
Mitigation: Simplified UI and guided QR-based verification

Scalability Risk

Risk: System not optimized for large-scale usage
Mitigation: Off-chain aggregation and batching strategies

Regulatory Risk

Risk: Lack of standardization in credential recognition
Mitigation: Align with global standards (e.g., W3C Verifiable Credentials)

9. Conclusion

The proposed impact measurement framework establishes a structured approach to evaluating both technical performance and real-world impact of the Digital Identity & Credential Verification system. By combining on-chain data, off-chain analytics, and institutional feedback, the framework enables continuous monitoring of adoption, efficiency, and trust outcomes throughout the pilot phase.

The system is expected to demonstrate measurable improvements in verification speed, reduction in manual processes, and increased trust in academic credentials, forming a strong foundation for future scale and integration.